

Ethanol Blending in Gasoline - Jamaica

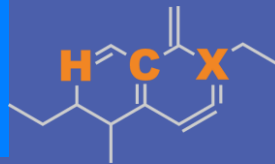
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**U.S. GRAINS &
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FARO90



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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



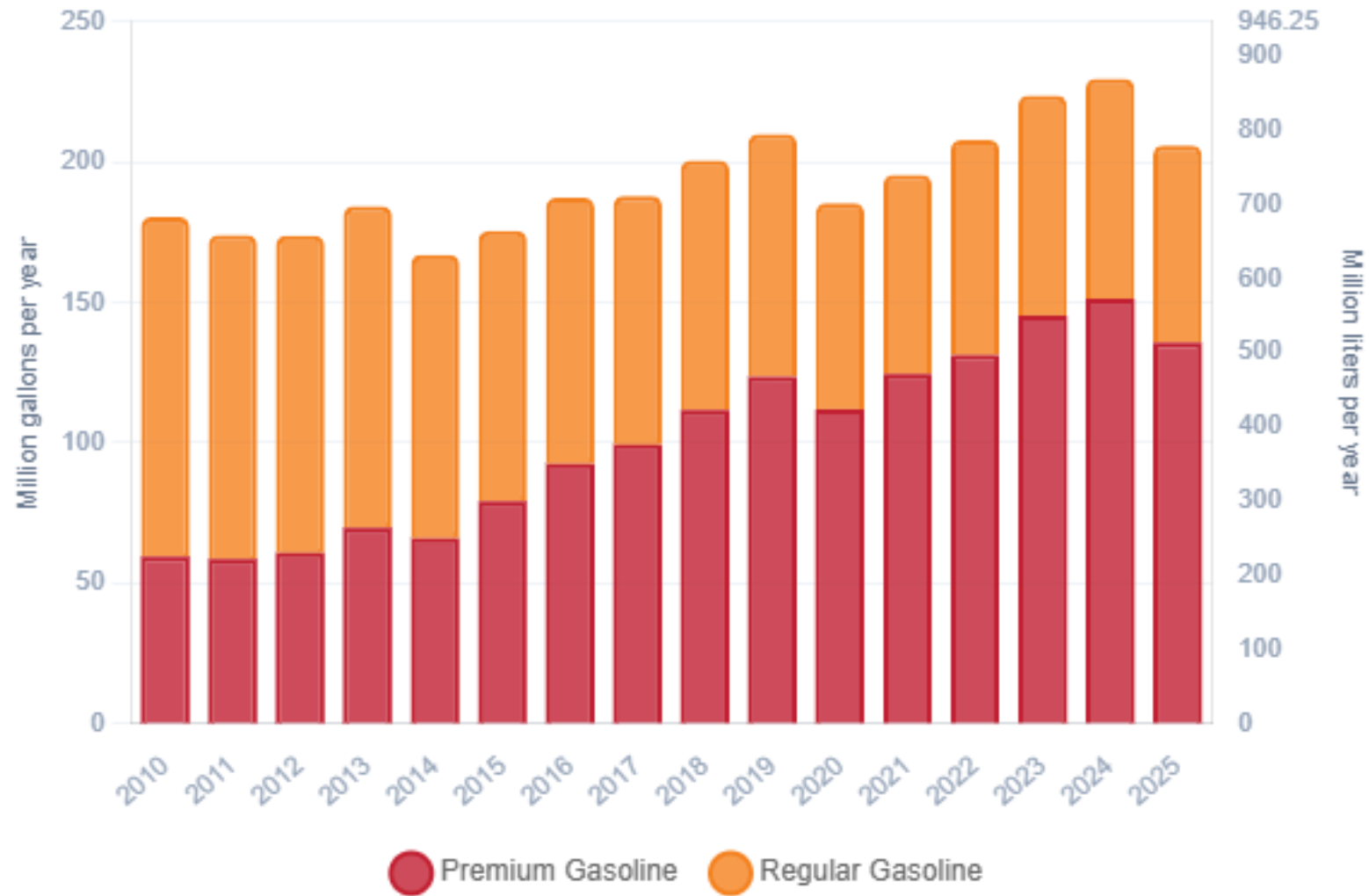
Ethanol Blending in Gasoline - Jamaica



In 2025, gasoline demand was 206 million gallons (779 million liters). Gasoline production was 34 million gallons (130 million liters) and imports 195 million gallons (739 million liters) mainly from United States. Regular grade (AKI 87) had a market share of 35% and Premium grade (AKI 90) 65%.

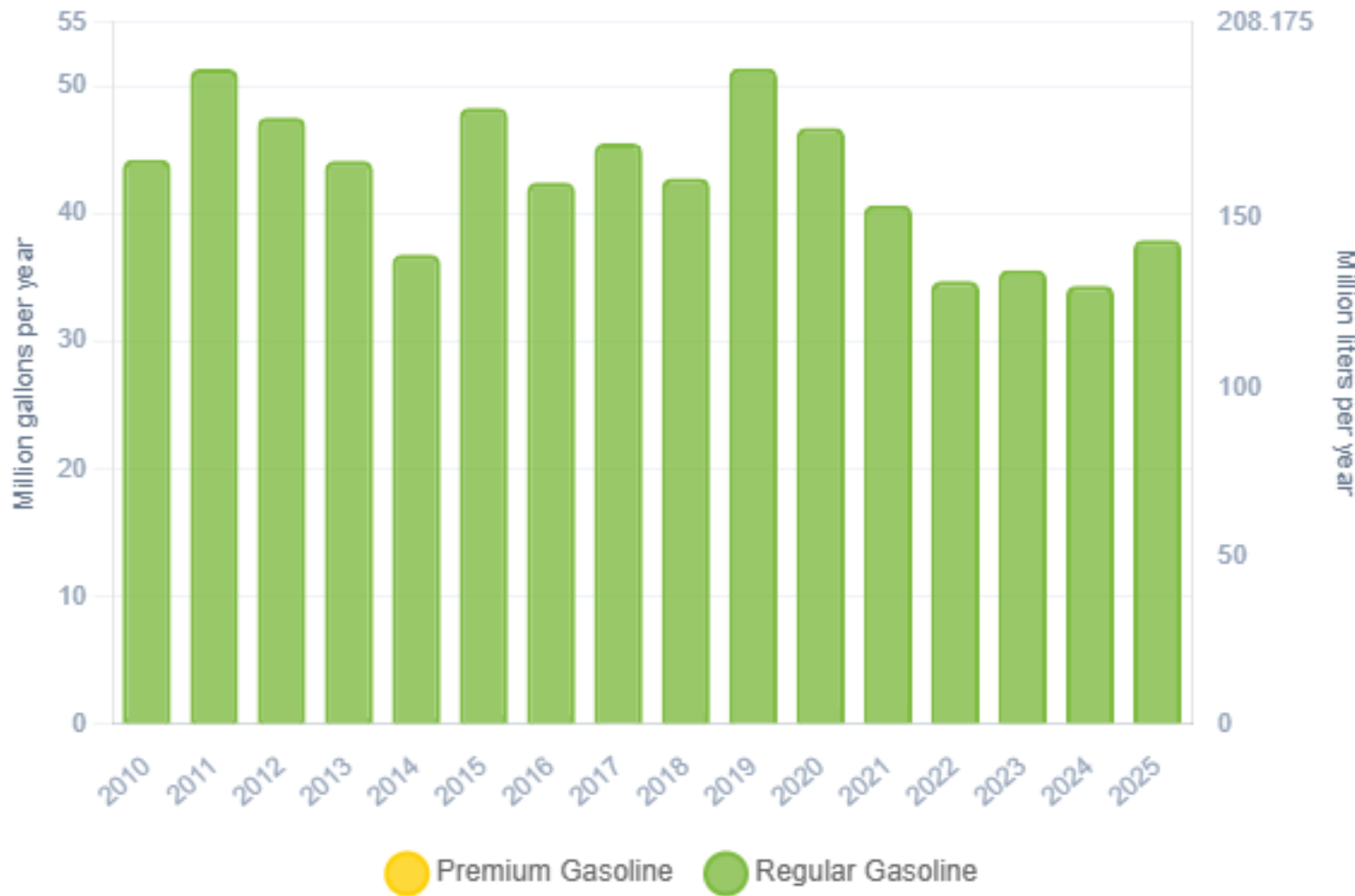
Current gasoline quality norm mandates E10 (JS 341:2017) . Previously, ethanol was supplied with local production to Petrojam, however, in recent years, ethanol is imported from the United States.

Gasoline Demand in Jamaica



Source: Petrojam, MSET

Gasoline Production in Jamaica

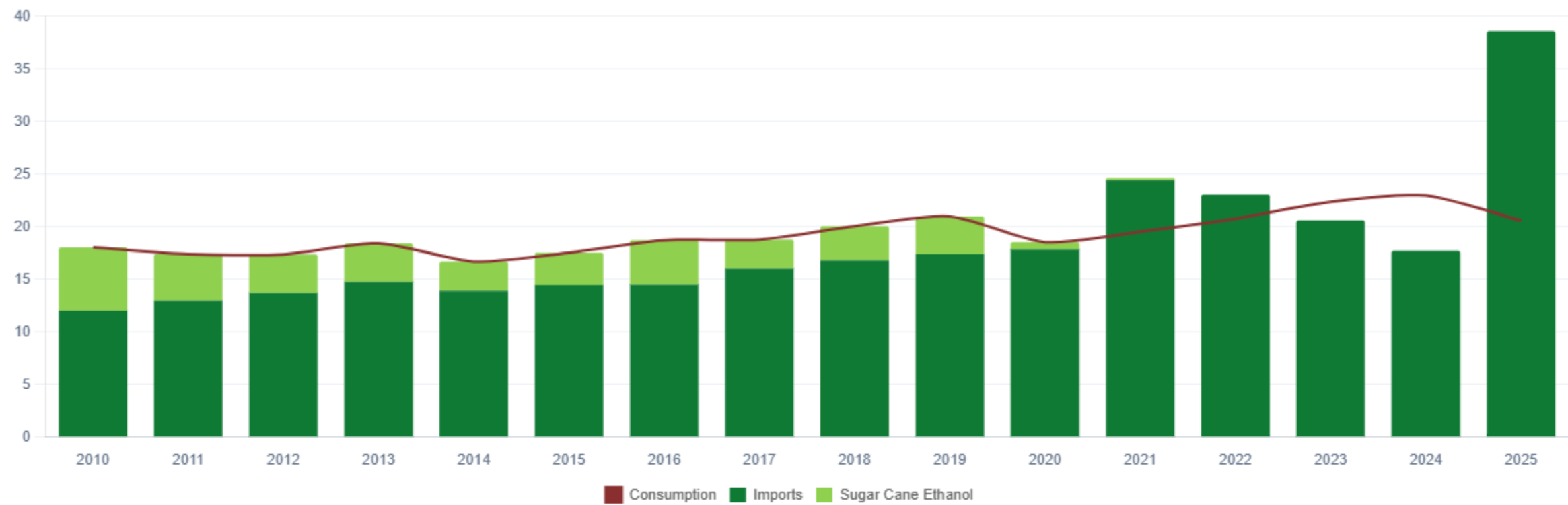


Source: Petrojam, MSET

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups at the 3 and 4 positions.



Ethanol Balance in Jamaica



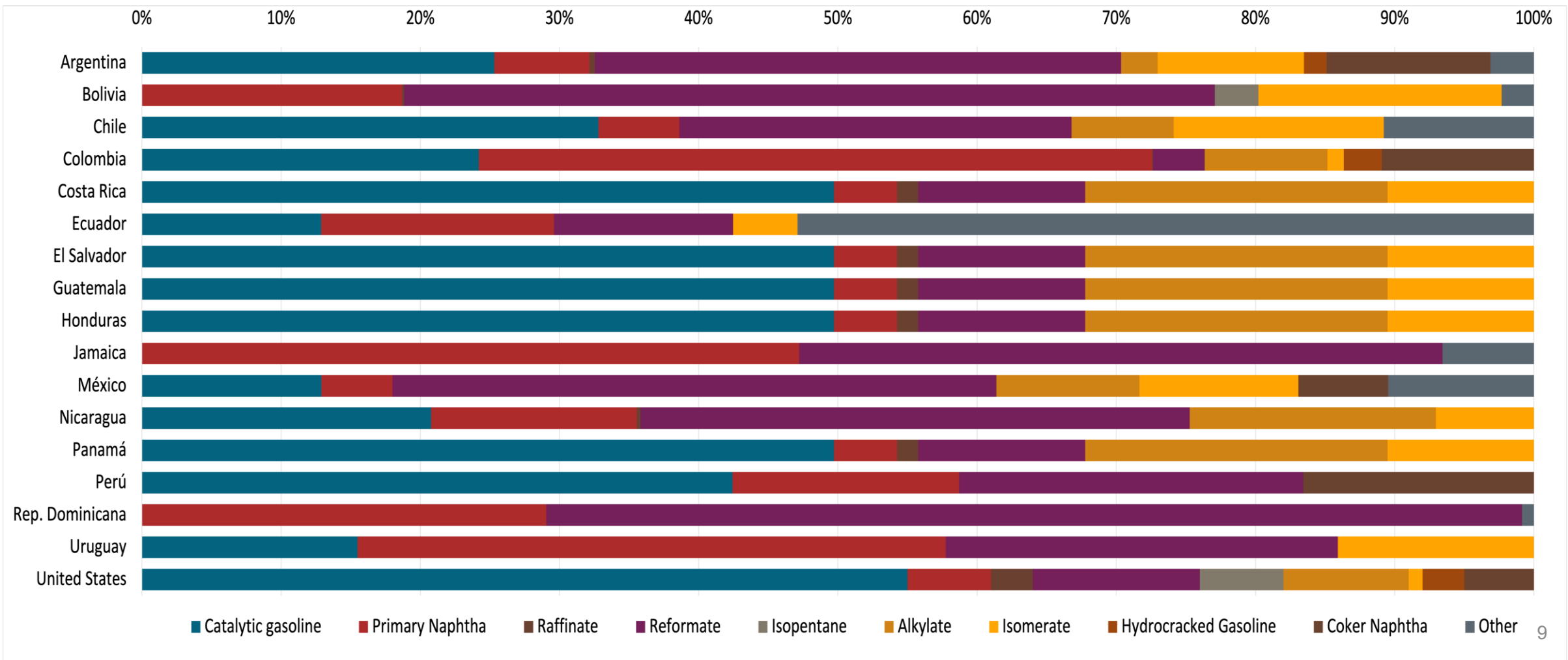
Source: MSET

Gasoline Quality in Jamaica

Name	JS 341:2017		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Unleaded Petrol AKI 87 E10	Unleaded Petrol AKI 90 E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5 %v/v	< 5 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45 %v/v	< 45 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON			> 95	> 95	> 98	> 98
MON			> 85	> 88	> 85	> 88
AKI	90	87				
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4 %m/m (Gasoline with no ethanol)	< 4 %m/m (Gasoline with no ethanol)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	< 15% v/v for Gasoline with no ethanol	< 15% v/v for Gasoline with no ethanol	-	-	-	-
Ethers 5 or more C Atoms	15% v/v (other oxygenates)	15% v/v (other oxygenates)	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



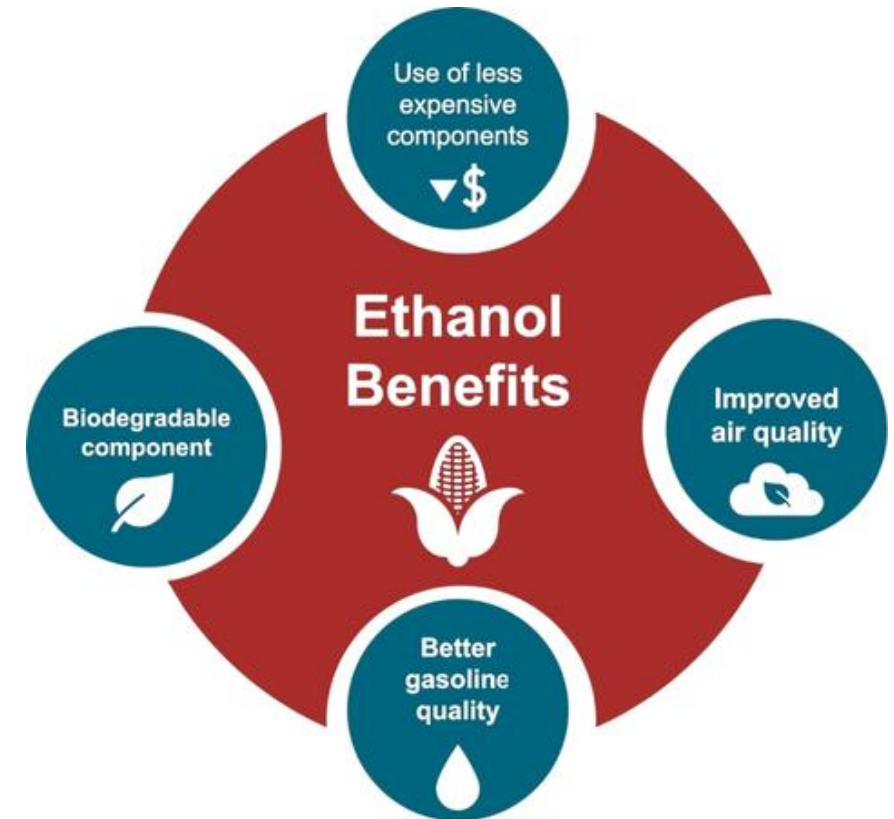
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

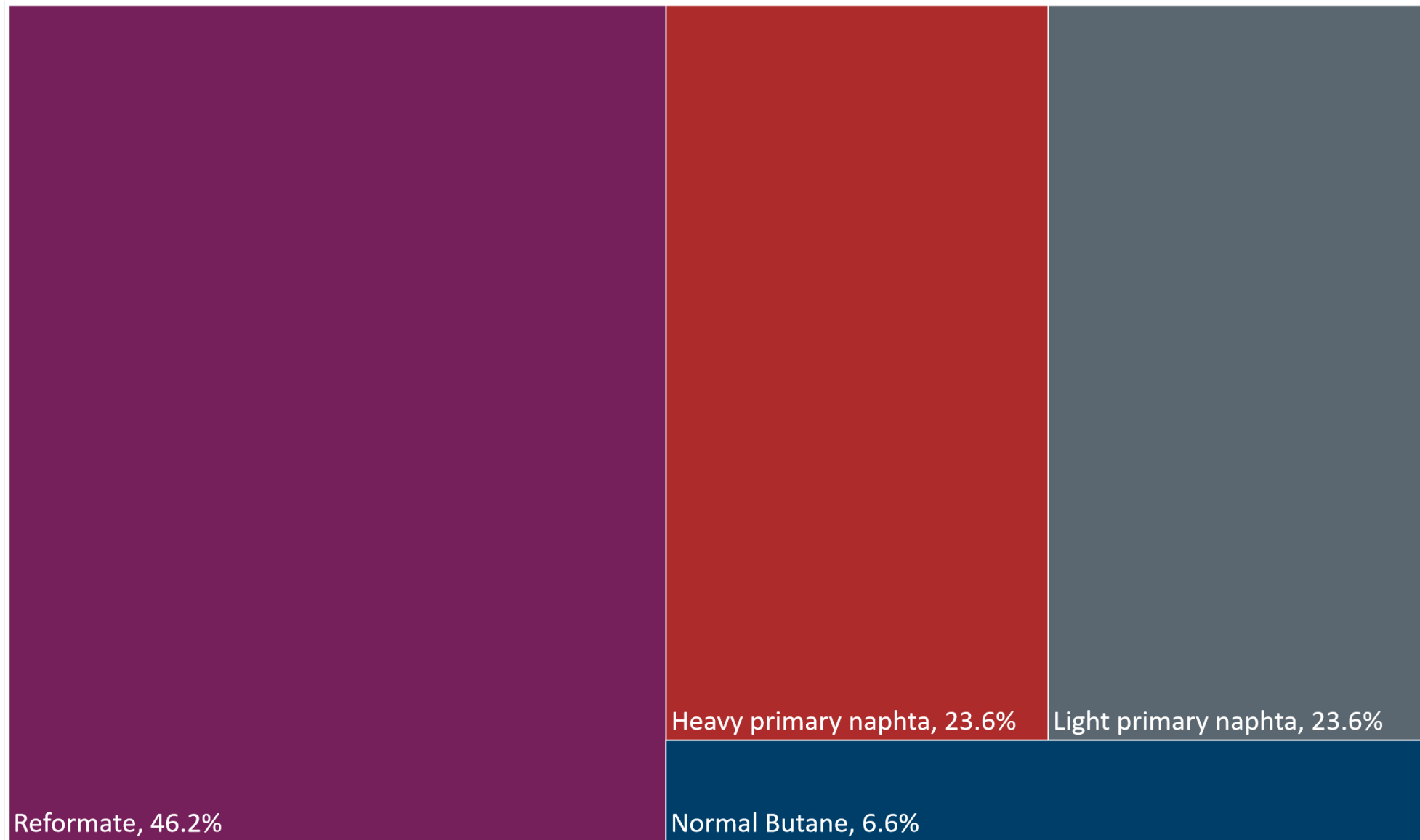
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

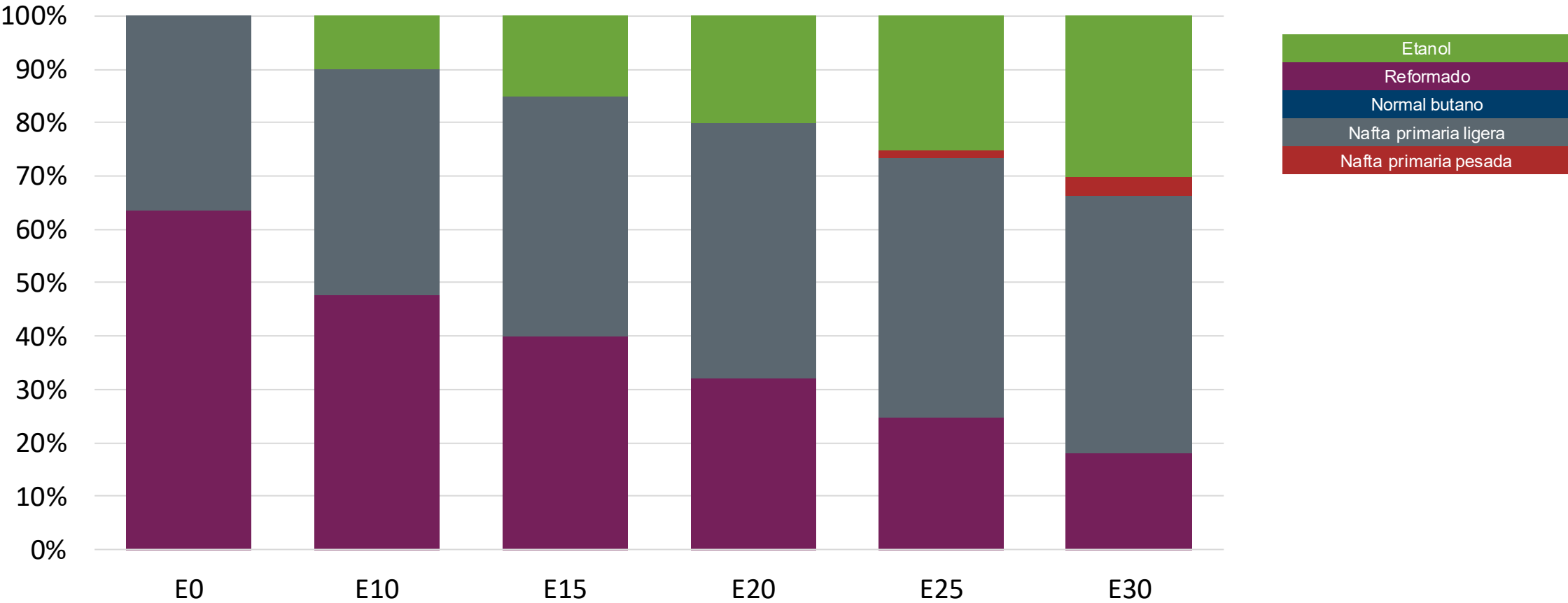
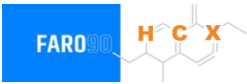
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components



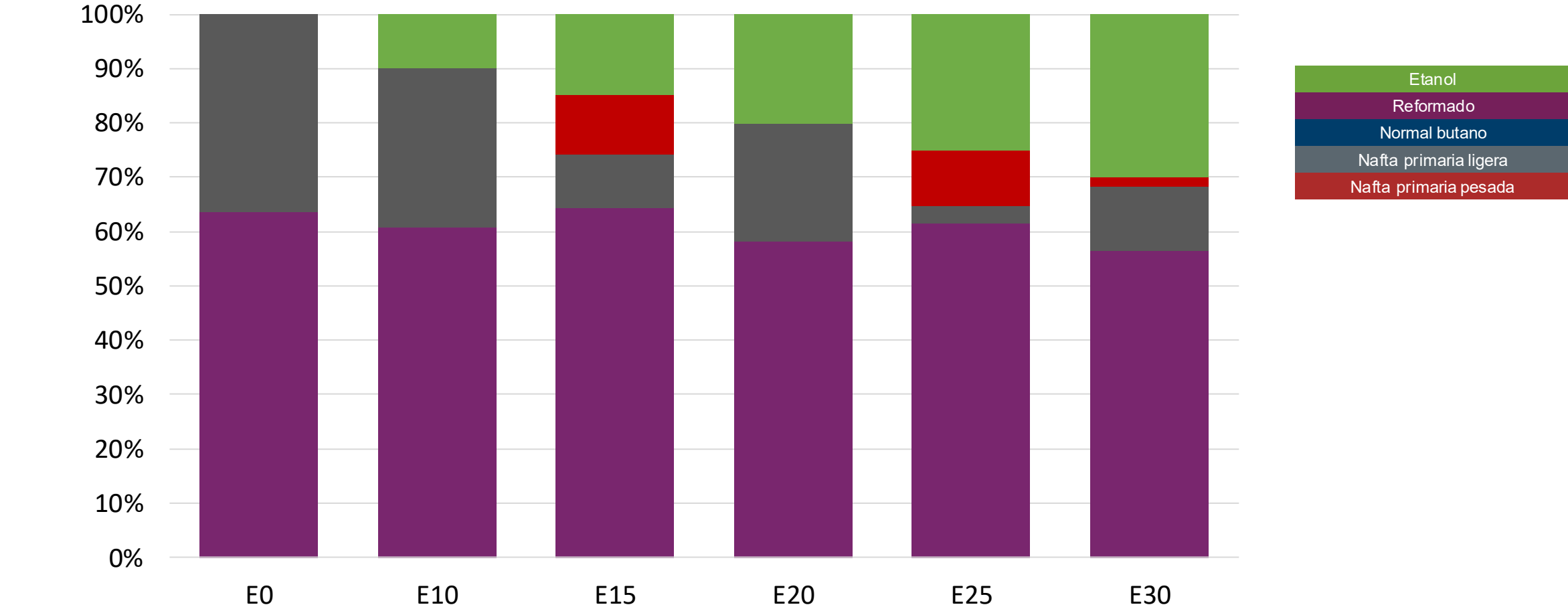
Regular Gasoline – Constant Octane



Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (USD/gal)	\$ 2.302	\$ 2.103	\$ 2.004	\$ 1.904	\$ 1.805	\$ 1.706

Source: Faro90

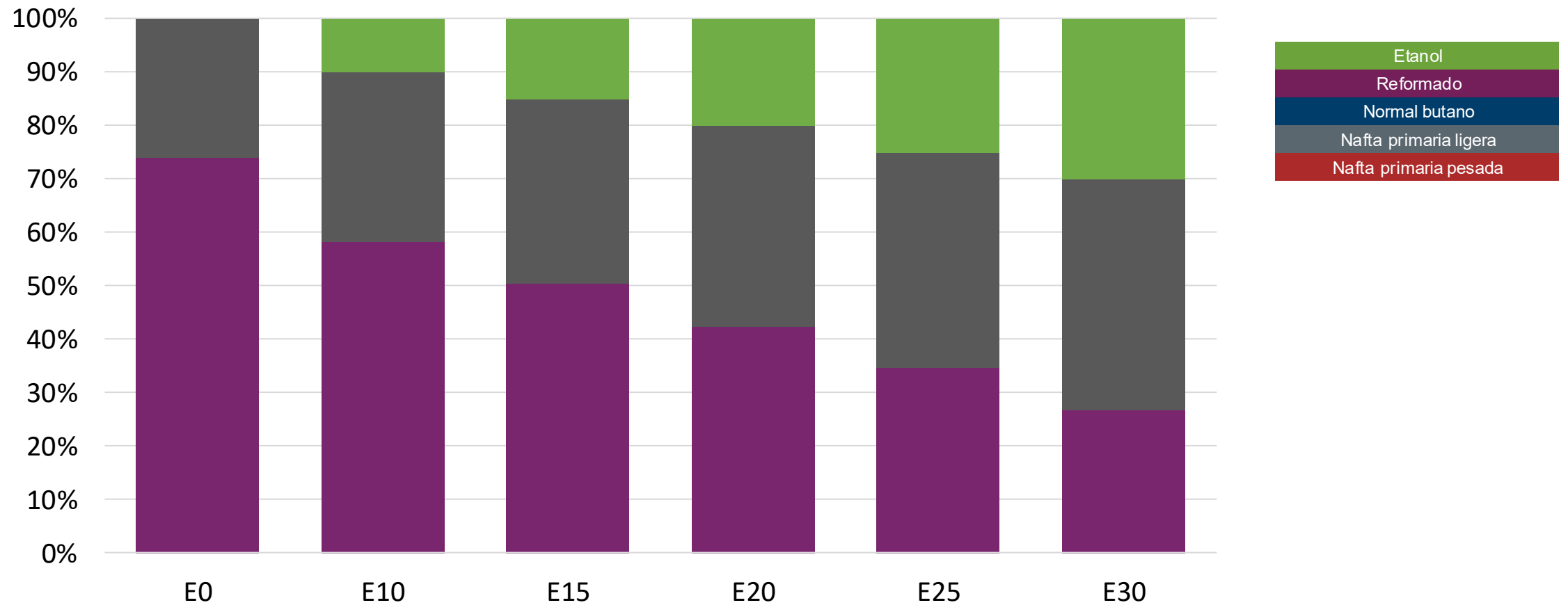
Regular Gasoline – Increased Octane



Octane (AKI)	87.0	90.8	92.6	94.5	96.4	98.3
Price (USD/gal)	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302

Source: Faro90

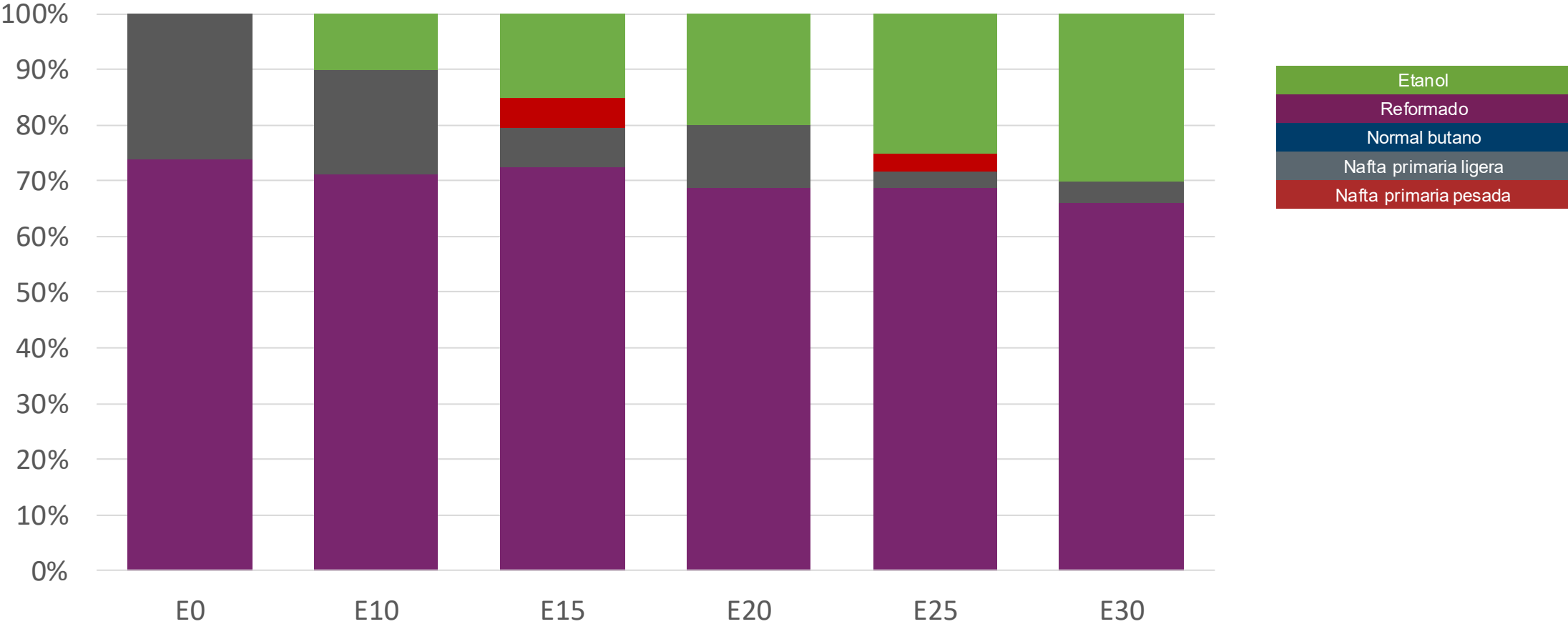
Premium Gasoline – Constant Octane



Octane (AKI)	90.0	90.0	90.0	90.0	90.0	90.0
Price (USD/gal)	\$ 2.460	\$ 2.261	\$ 2.162	\$ 2.062	\$ 1.963	\$ 1.864

Source: Faro90

Premium Gasoline – Octane Increment



Octane (AKI)	90.0	93.8	95.6	97.5	99.4	101.3
Price (USD/gal)	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

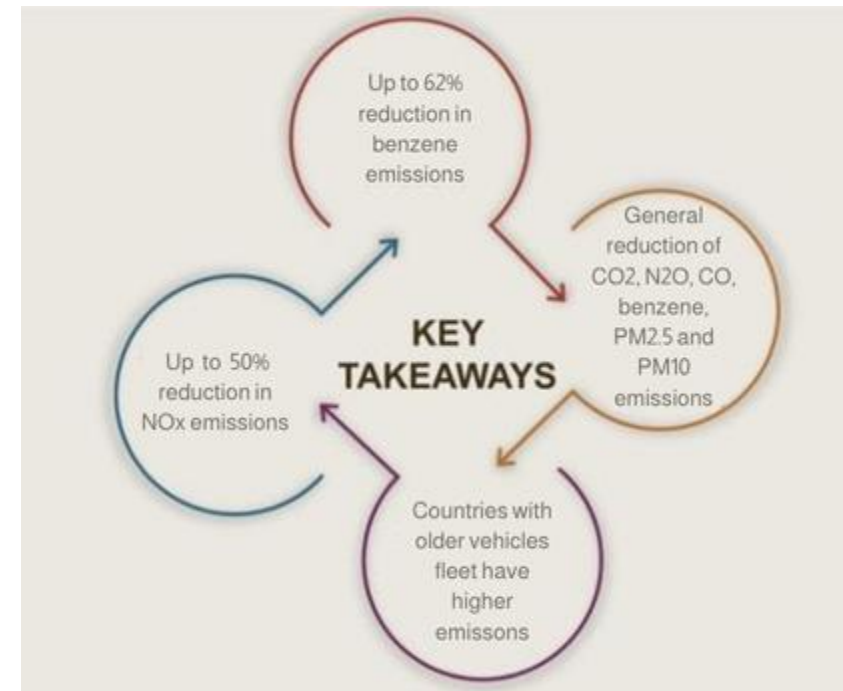
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Jamaica

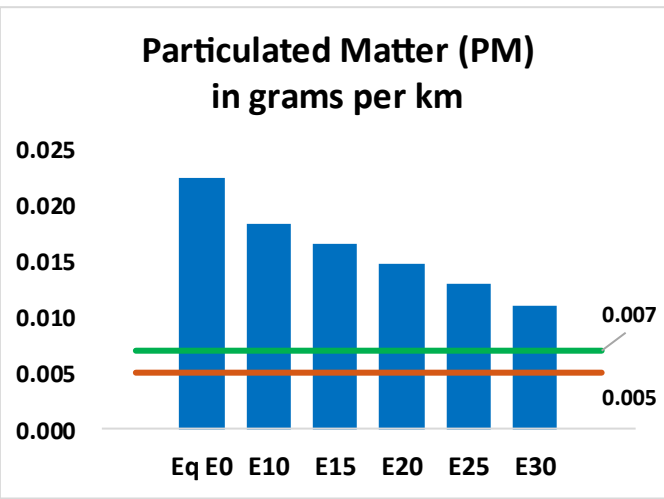
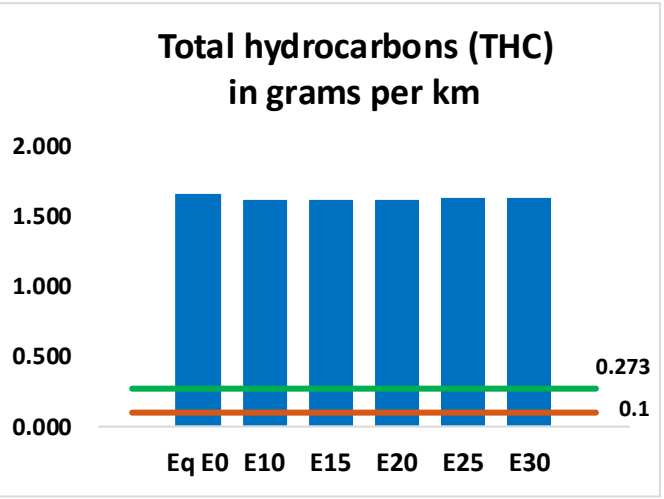
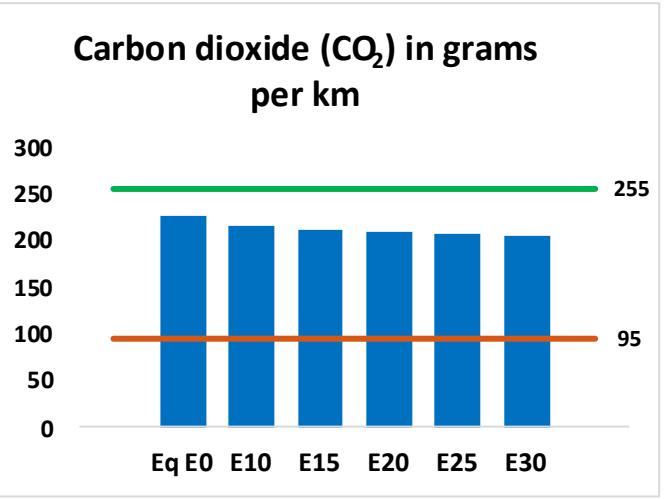
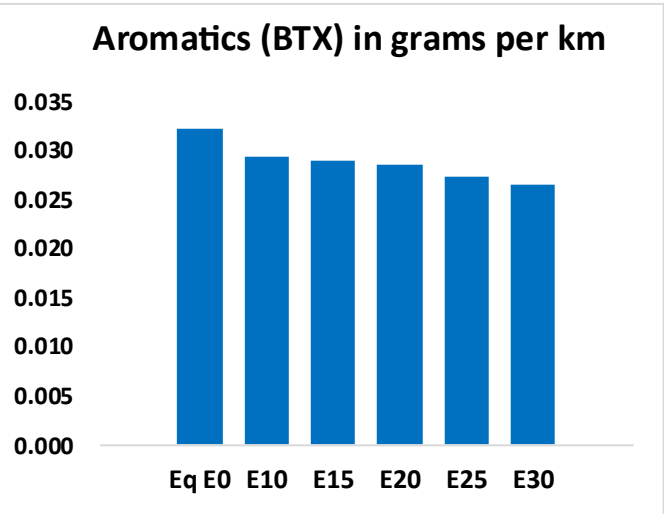
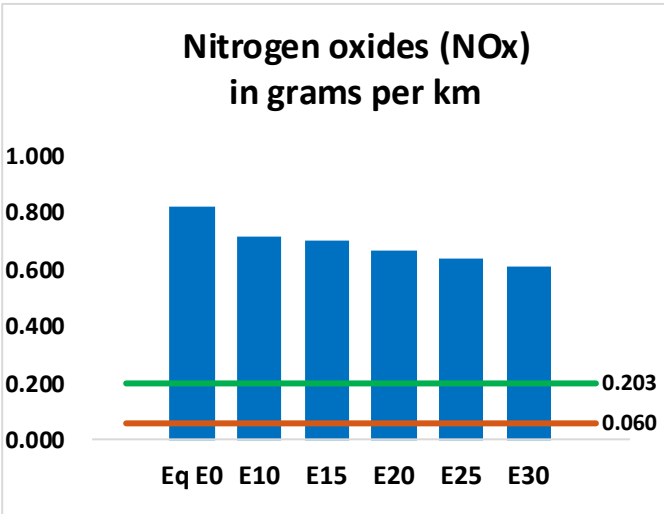
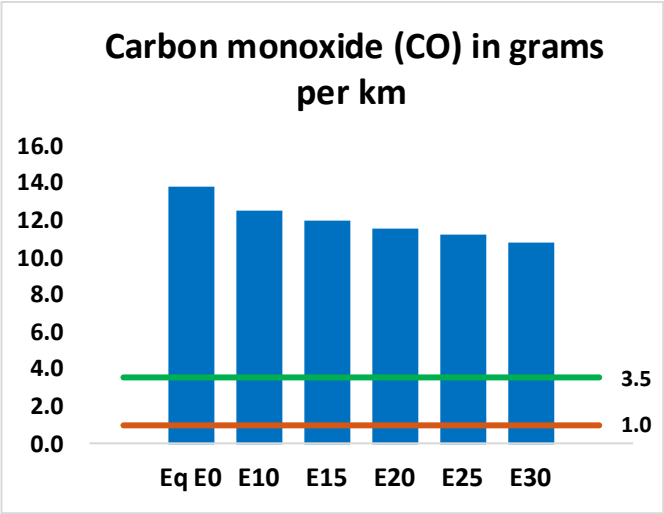
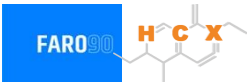
Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles							
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	4,644							
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	15,718							
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	114,014							
		Euro III		> 8 years, > 160 tkm	79,316							
		Euro IV		> 8 years, > 160 tkm	57,226							
		Euro V / Euro 6		> 8 years, > 160 tkm	42,786							
				4-8 years, 80-160 tkm	12,716							
				< 4 years, < 80 tkm	9,435							
	Carburator	3-Ways		PCV Tank	> 8 years, > 160 tkm	8,366						
	Single FI-Pt	3-Ways			> 8 years, > 160 tkm	76,028						
Medium automobiles	Multi FI-Pt	3-Ways	PCV Tank	> 8 years, > 160 tkm	169,895							
		Euro III		> 8 years, > 160 tkm	75,299							
		Euro IV		> 8 years, > 160 tkm	46,074							
		Euro V / Euro 6		> 8 years, > 160 tkm	45,043							
				4-8 years, 80-160 tkm	29,720							
				< 4 years, < 80 tkm	50,123							
		Small Engines		FI 4-ciclos	None	PCV	> 8 years, > 160 tkm	682				
					Euro II		> 8 years, > 160 tkm	841				
Euro III	> 8 years, > 160 tkm		8,671									
Euro IV	> 8 years, > 160 tkm		3,974									
Euro V	4-8 years, 80-160 tkm		5,269									
	4-8 years, 80-160 tkm		6,936									
	< 4 years, < 80 tkm		22,972									
Carburator	3-Ways		PCV	> 8 years, > 160 tkm	1,135							
FI	3-Ways			> 8 years, > 160 tkm	16,622							
	Euro III			> 8 years, > 160 tkm	5,579							
	Euro IV	> 8 years, > 160 tkm		2,778								
	Euro V / Euro 6	> 8 years, > 160 tkm		2,170								
		4-8 years, 80-160 tkm		1,985								
		< 4 years, < 80 tkm		4,520								
	Heavy	Hybrid			Any							
		Electric			Any							

Vehicular Fleet: **885,746**
 Average Age: **20 years**
 Motorcycles: **2.3%**

Source: MSET, Faro 90

Jamaica – Vehicular Emissions

United States
Euro 6

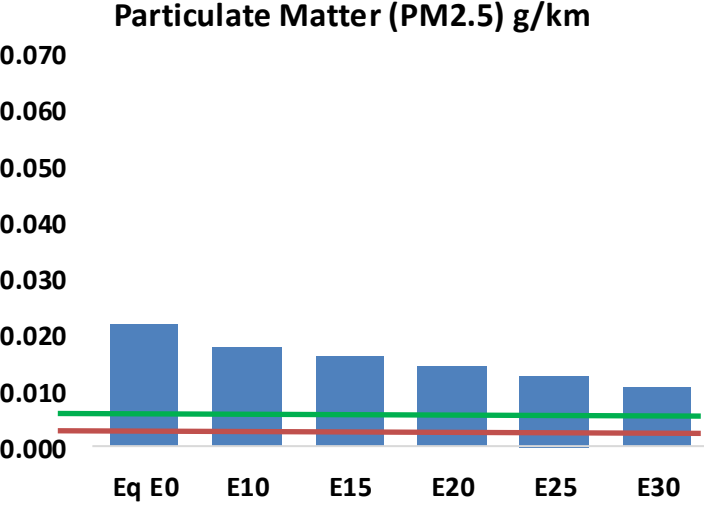
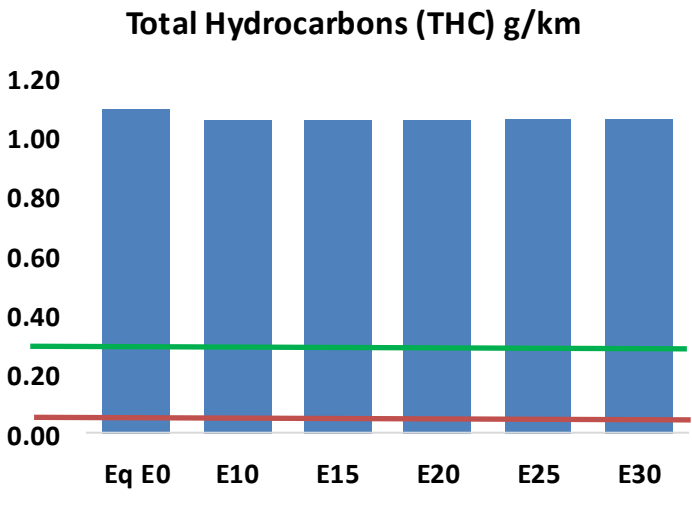
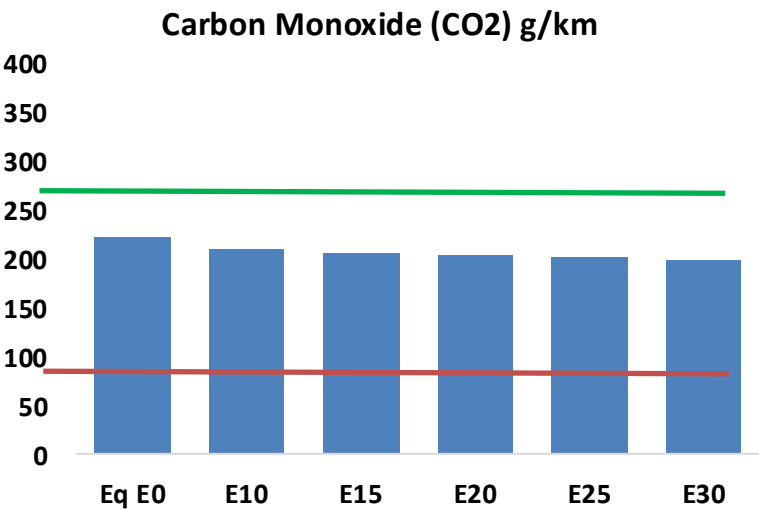
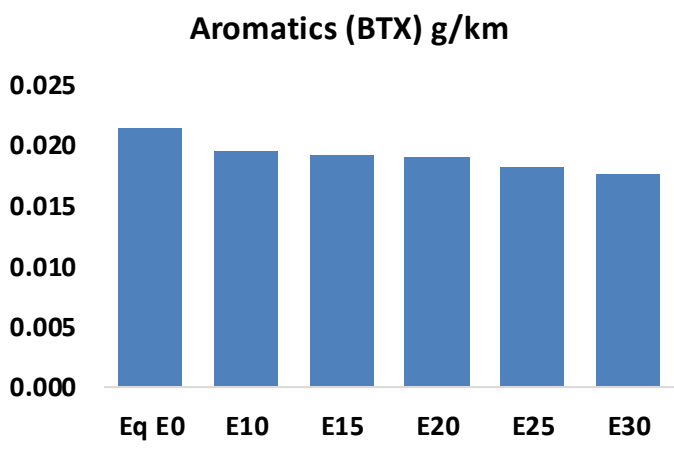
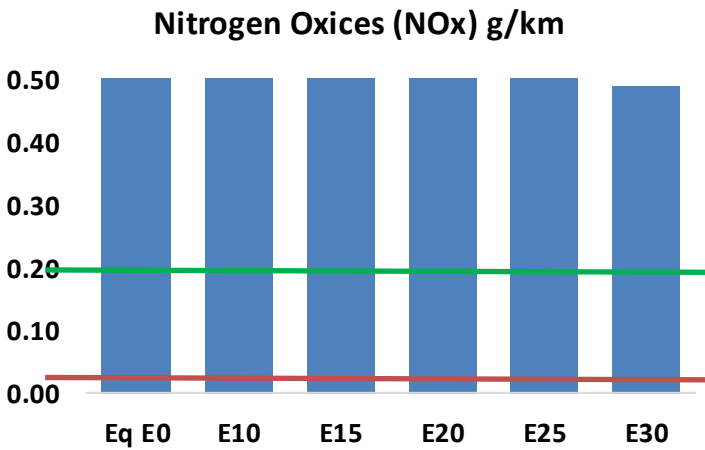
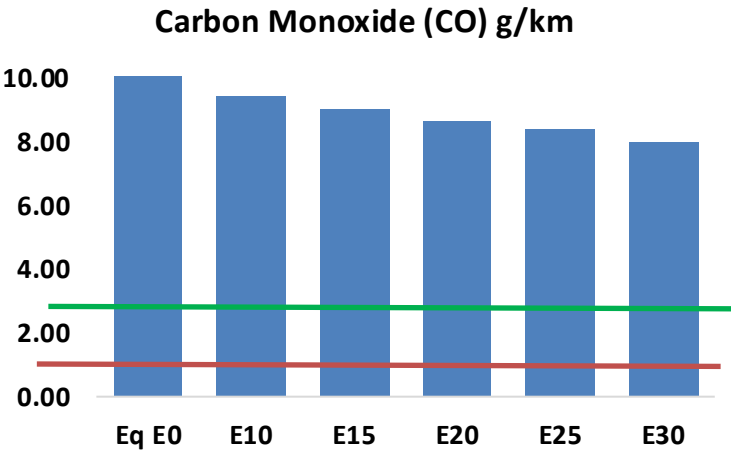
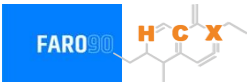


Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	314,970	299,702	284,433	273,667	263,636	256,165	246,058	-10%	-16%
CO2	5,175,847	5,042,754	4,917,055	4,818,196	4,769,277	4,747,659	4,660,146	-5%	-8%
VOC	37,858	37,303	36,748	36,752	36,968	37,198	37,280	-3%	-2%
NOx	18,864	17,418	16,459	15,989	15,211	14,610	13,897	-13%	-19%
SOx	62	57	52	48	45	40	36	-15%	-28%
NH3	1,724	1,707	1,690	1,698	1,717	1,731	1,742	-2%	0%
N2O	93	93	92	93	95	96	97	-1%	2%
CH4	8,151	8,151	8,151	8,274	8,453	8,608	8,755	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	132	127	122	118	116	114	111	-8%	-12%
Acetaldehyde	231	265	388	592	806	921	1,088	68%	249%
Formaldehyde	668	649	757	889	931	1,030	1,122	13%	39%
Benzene	736	708	670	660	654	626	606	-9%	-11%
PM2.5	285	259	233	210	188	164	139	-18%	-34%
PM10	230	209	188	169	151	132	112	-18%	-34%

Jamaica – Vehicular Emissions

United States
Euro 6



Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	166,859	158,208	149,556	143,282	137,342	132,889	127,019	-10%	-18%
CO2	3,490,486	3,400,730	3,315,961	3,249,293	3,216,303	3,199,172	3,140,202	-5%	-8%
VOC	17,388	17,100	16,813	16,782	16,845	16,922	16,924	-3%	-3%
NOx	10,587	9,775	9,234	8,968	8,528	8,188	7,785	-13%	-19%
SOx	41	38	35	32	29	27	24	-15%	-28%
NH3	1,181	1,169	1,158	1,163	1,176	1,186	1,193	-2%	0%
N2O	54	54	54	54	55	56	56	-1%	2%
CH4	3,735	3,735	3,735	3,791	3,873	3,944	4,011	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	95	92	89	87	86	86	84	-6%	-9%
Acetaldehyde	175	200	294	448	610	697	824	68%	249%
Formaldehyde	501	487	568	667	698	773	841	13%	39%
Benzene	339	326	308	304	301	288	279	-9%	-11%
PM2.5	200	181	163	147	132	115	97	-18%	-34%
PM10	147	134	120	109	97	85	72	-18%	-34%

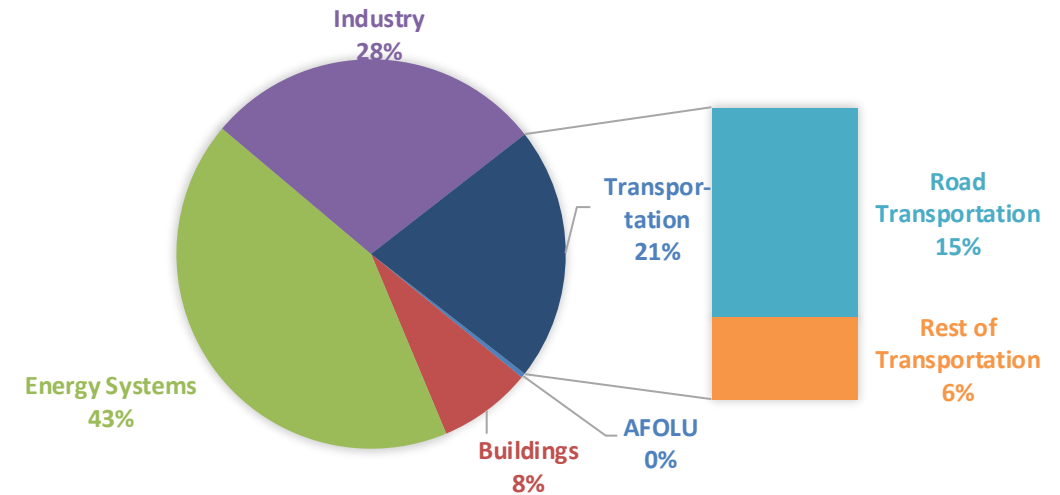
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

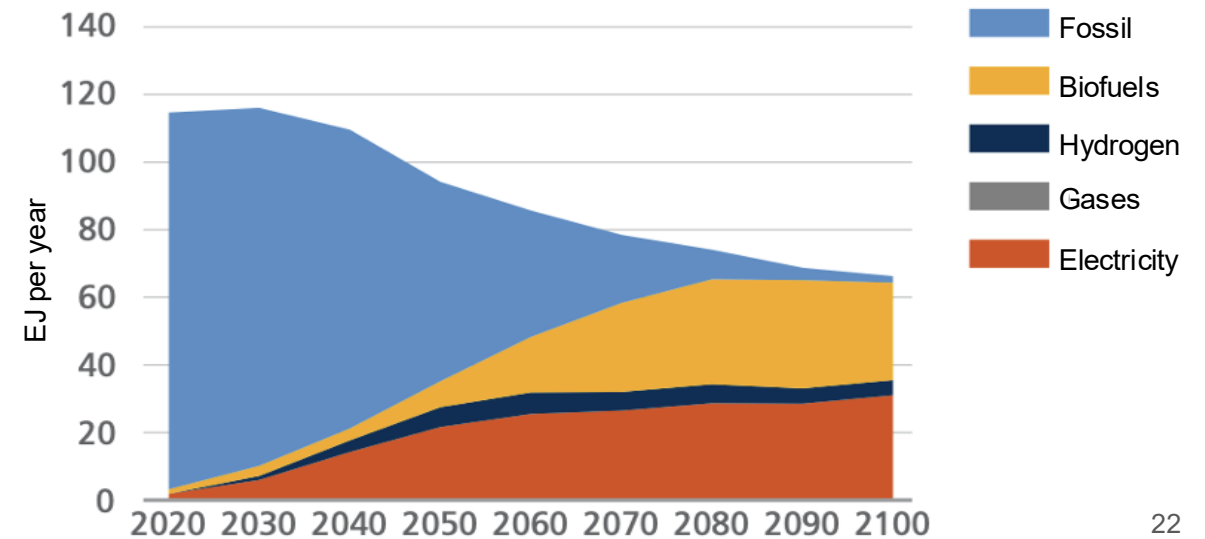
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

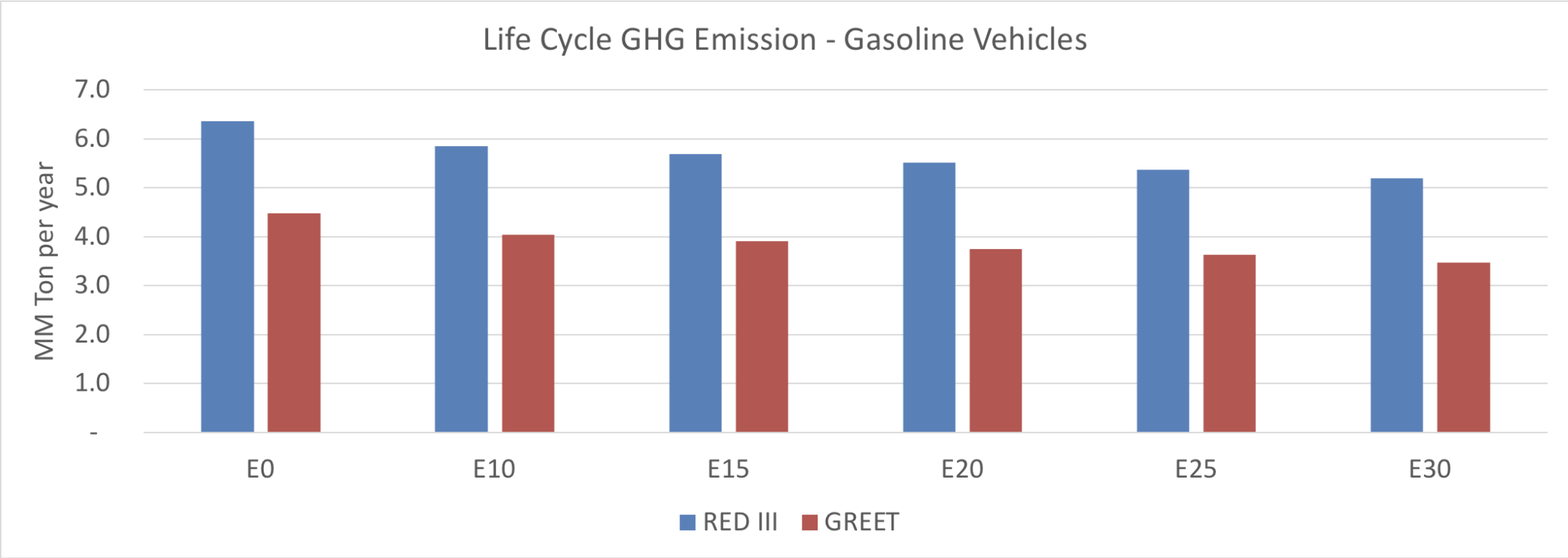
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Jamaica – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	9.4
2035 Target (MMT/year)	4.7
Delta	4.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.9%	16.2%	18.9%	22.5%
RED II/III	0.0%	7.9%	10.5%	13.2%	15.5%	18.4%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.2%	15.4%	17.9%	21.3%
RED II/III	0.0%	10.7%	14.1%	17.8%	20.9%	24.8%